

GIVE REASONS

Blood Physiology

Decrease in RBC count pregnancy

- In pregnancy, the RBC count decreases. It is because of increase in ECF volume. Increase in ECF volume, increases the plasma volume also resulting in hemodilution. So, there is a relative reduction in the RBC count.

Temporary increase in RBC count after exercise

- There is a temporary increase in RBC count after exercise. It is because of mild hypoxia and contraction of spleen. Spleen stores RBCs. Hypoxia increases the sympathetic activity resulting in secretion of adrenaline from adrenal medulla. Adrenaline contracts spleen and RBCs are released into blood

Inhabitants of mountains (above 10,000 feet from mean sea level) have an increased RBC count of more than 7 million/cu mm.

- It is due to hypoxia (decreased oxygen to tissues) in high altitude because low oxygen tension Hypoxia stimulates kidney to secrete a hormone called erythropoietin which stimulate the bone marrow to produce more RBCs

Repeated radiation or some chemicals leads to Aplastic anemia

- Aplastic anemia is anemia due to destruction of the bone marrow, which may be caused by chemicals (including benzene and arsenic) or by radiation. Repeated exposure of ionising radiation or pollutants may damage the blood bone marrow and is replaced by yellow bonemarrow leads to decreased erythroopoesis

Gastrectomy causes pernicious anemia

- In pernicious anemia there is an inadequate amount of vitamin B12, which is needed for red blood cell production. The glandular gastric mucosa of the stomach, which normally secretes a protein called intrinsic factor. In the absence of intrinsic factor, the vitamin B12 obtained in the diet cannot be absorbed by intestinal cells.

No ABO incompatibility

- Normally ABO blood groups are IgM class of immunoglobulin, cannot cross the placenta
- The only group of immunoglobulin that cross placenta is IgG type (erythroblastosis foetalis)

Edema due to decreased amount of plasma proteins

- When the amount of plasma proteins decreases, the colloidal osmotic pressure decreases. Because of this, the permeability of the capillary increases, resulting in increased capillary filtration. So, more amount of water leaks out of the capillary. It accumulates in the tissue spaces resulting in extracellular edema

Muscle Physiology

Stiffness of muscles and joints after death

- Rigor mortis Soon after death, the cell membrane becomes highly permeable to calcium. So a large number of calcium ions enters the muscle fibers and promotes the formation of actomyosin complex resulting in contraction of the muscles. It is also triggered by depletion of ATP and creatinin phosphate. Few hours after death, all the muscles of body undergo severe contraction and become rigid. The joints also become stiff and locked

A disease characterized by progressive degeneration of muscle fibers, without the involvement of nervous system

- Duchenne muscular dystrophy is a sexlinked recessive disorder. It is due to the absence of a gene product called dystrophin in the X chromosome. Dystrophin is necessary for the stability of sarcolemma. This disease is characterized by degeneration and necrosis of muscle fibers. The degenerated muscle fibers are replaced by fat and fibrous tissue
- Becker muscular dystrophy is also a sexlinked disorder. It occurs due to the reduction in quantity or alteration of dystrophin

An autoimmune disease of neuromuscular junction caused by antibodies

- Myasthenia gravis is caused due to the development of autoanti bodies (IgG autoantibodies) against the receptors of acetylcholine. That is, the body develops antibodies against its own acetylcholine receptors. These antibodies prevent binding of acetylcholine with its receptors or destroy the receptors. So, though the acetylcholine release is normal, it cannot execute its action
- Muscles which are more susceptible for myasthenia gravis are muscles of neck, limbs, eyeballs and the muscle responsible for eyelid movements, chewing, swallowing, speech and respiration

Lambert Eaton syndrome

- A disorder of neuromuscular junction caused by development of antibodies against calcium channel in the nerve terminal, resulting in reduction in the release of quanta of acetylcholine. So no endplate potential and no muscular transmission. This disease is commonly associated with carcinoma. So, it is also called carcinomatous myopathy

Digestive System

The yellow coloration of the skin, mucous membrane and deeper tissues

- Jaundice or icterus is the condition characterized by yellow coloration of the skin, mucous membrane and deeper tissues due to increased bilirubin level in blood. The normal serum bilirubin level is 0.5 to 1.5 mg/dL. Jaundice occurs when bilirubin level exceeds 2 mg/dL.
1. Prehepatic or hemolytic jaundice

2. Hepatic or hepatocellular jaundice
3. Posthepatic or obstructive jaundice.

The formation of bulky, foul smelling, frothy and clay coloured stools with large quantity of undigested fat

- Steatorrhea - lack of pancreatic lipase causes impaired digestion and absorption of fat
- Absence of bile salts - Bile salts are essential for the digestion of fat by lipase and absorption of fat from intestine

Rapid defecation in children after intake of food

- Gastrocolic reflex - it is physiological reflex that control the mobility of lower gastrointestinal tract, as a result of gastrocolic reflex the colon has increased mobility in response to the stretch of the stomach with ingestion of food, which allows more space for food and create movements of ingested food towards rectum

Renal Physiology

Renal failure leads to death

- Primary function of kidneys is homeostasis. It is accomplished by the formation of urine. During the formation of urine, kidneys regulate various activities in the body, which are concerned with homeostasis such as excretion of Waste Products, maintenance of Water Balance, maintenance of Electrolyte Balance and maintenance of Acid–Base Balance Failure this homeostasis leads to death

Chronic renal disease produces anemia

- Erythropoetin a hormone produced by kidney which is the stimulating factor to RBC formation

Polyurea in diabetes mellitus

- Polyuria is the increased urinary output due to osmotic diuresis. Osmotic diuresis is the diuresis induced by the osmotic effects of glucose in urine which increases the osmotic pressure in the tubular fluid thereby water is diffused into it leads to less reabsorption of water and more urine output

Polyurea in diabetes insipidus

- The renal tubules fail to reabsorb water because of ADH deficiency

Endocrinology

Excess growth of the body in children

- Gigantism - Hypersecretion of GH in childhood or in pre-adult life before the fusion of epiphysis of bone with shaft. . Hypersecretion of GH is because of tumor of acidophil cells in the anterior pituitary

Excess growth of the body in adults

- Acromegaly is due to hypersecretion of GH in adults after the fusion of epiphysis with shaft of the bone. Hypersecretion of GH is because of tumor of acidophil cells in the anterior pituitary gland

Obesity with Fatty hump between the shoulders, a rounded face and pink or purple stretch marks

- Cushing disease develops by basophilic adenoma of adenohypophysis. It increases the secretion of adrenocorticotrophic hormone, which in turn stimulates the adrenal cortex to release cortisol. Cushing disease also develops by hyperplasia or tumor of adrenal cortex.

A 30- years-old person looks like a 60-years-old person with dry and wrinkled skin

- Simmond disease is a rare pituitary disease(pituitary cachexia) - hyposecretion of all the anterior pituitary hormones(panhypopituitarism) due to the atrophy or degeneration of anterior pituitary

A rare disease in adults characterized by the atrophy of the extremities of the body

- Acromicria - Deficiency of GH in adults

Obesity is accompanied by sexual infantilism mainly the adolescent boys

- Dystrophia adiposogenitalis (Fröhlich syndrome or hypothalamic eunuchism) - Dystrophia adiposogenitalis is due to hypoactivity of both anterior pituitary and posterior pituitary. Common cause of this disease is the tumor in pituitary gland and hypothalamic regions, concerned with food intake and gonadal development. Other causes are injury or atrophy of pituitary gland and genetic inability of hypothalamus to secrete luteinizing hormone-releasing hormone

Hypocalcemic tetany after thyroidectomy

- Thyroidectomy is the removal thyroid gland, during surgery accidental removal injury or devascularisation of parathyroid gland which secrete parathormone. Parathormone regulate calcium level by enhancing the resorption of calcium from the bones (osteoclastic activity) by acting on osteoblasts and osteoclasts of the bone, PTH increases the reabsorption of calcium from the renal tubules along with magnesium ions and hydrogen ions. It increases calcium reabsorption mainly from distal convoluted tubule and proximal part of collecting duct and it increases the absorption of calcium ions from the GI tract indirectly. Hypocalcemia causes neuromuscular hyperexcitability, resulting in hypocalcemic tetany.

Polyurea, polydipsia and polyphagia in diabetes mellitus

- Excess urine formation with increase in the frequency of voiding urine is called polyuria. It is due to the osmotic diuresis caused by increase in blood glucose level.

- Polydipsia Increase in water intake. Excess loss of water decreases the water content and increases the salt content in the body and loss of water which reduces ECF volume. This stimulates the thirst center in hypothalamus. Thirst center, in turn increases the intake of water.
- Polyphagia means the intake of excess food. Glucose is not utilised by the cell due to insulin deficiency or resistance so energy production is very low which stimulate feeding centre of hypothalamus.

Deficiency or low level of aldosterone is fatal

- Aldosterone is very essential for life and it maintains the osmolarity and ECF volume. It is usually called life-saving hormone because, its absence causes death within 3 days to 2 weeks. Aldosterone mainly helps the reabsorption of sodium from renal tubules, excretion of potassium through renal tubules and secretion of hydrogen into renal tubules.

Exposure to even mild harmful stress after adrenalectomy, leads to collapse and death.

- Removal of adrenal glands in human beings and animals causes disturbances of metabolism
Cortisol is a life-protecting hormone because, it helps to withstand the stress and trauma in life

Reproductive System

Mumps in males after puberty leads to sterility

- Blood-testes barrier is commonly damaged by trauma or viral infection like mumps. Whenever, the bloodtestes barrier is damaged the sperms enter the blood. The immune system of the body is activated, resulting in the production of autoantibodies against sperms. The antibodies destroy the germ cells, leading to consequent sterility

Osteoporosis in females after menopause

- After menopause the estrogen is not secreted or it becomes scanty. Estrogen increases osteoblastic activity. The lack of estrogen leads to osteoporosis, in which the bones become extremely weak and fragile. Because of this, the bones are highly susceptible for fractures

Menstrual periods in females

- Every month the lining of uterine wall thickens(poliferative phase) and egg is released from one of ovaries(ovulation). If pregnancy does not occurs the level of estrogen and progesterone decreases so the thickened endometrium is shed or desquamated. This desquamated endometrium is expelled out through vagina along with blood and tissue fluid. The process of shedding and exit of uterine lining along with blood and fluid is called menstruation or menstrual bleeding.

Oral contraceptives prevent pregnancy

- Use of contraceptive increases large amount of estrogen and progesterone in the blood which suppresses the release of gonadotropins, FSH and LH from pituitary by means of feedback mechanism. Lack of FSH and LH prevents the maturation of follicle, and ovulation. In addition, progesterone increases the thickness of mucosa in cervix, which is not favorable for transport of sperm. Progesterone inhibits the release of LH from hypothalamus through feedback effect. This effect is utilized for its contraceptive action.

Cardiovascular System

Increased stroke volume during exercise

- It is due to skeletal muscle pump, there is increase in venous return leading to increase in stroke volume (cardiac output is directly proportional to venous return)

Heart rate increases during inspiration and decreases during expiration (respiratory sinus arrhythmia)

- Sinus arrhythmia is due to the alteration of vagal tone because of impulses arising from respiratory centers during inspiration. These impulses inhibit the vasodilator area, resulting in decreased vagal tone and increased heart rate. During expiration, the respiratory center stops sending impulses to vasodilator center, vagal tone increases, leading to decrease in heart rate.

First heart sound

- Closure of atrioventricular valves

Second heart sound

- Closure of semilunar valves

Third heart sound

- Rushing of blood into ventricle

Fourth heart sound

- Contraction of atrial musculature

'P' wave in ECG

- P' wave is produced due to the depolarization of atrial musculature. Depolarization spreads from SA node to all parts of atrial musculature

'QRS' complex in ECG

- 'QRS' complex is due to depolarization of ventricular musculature. 'Q' wave is due to the depolarization of basal portion of interventricular septum. 'R' wave is due to the depolarization of apical portion of interventricular septum and apical portion of ventricular muscle. 'S' wave is due to the depolarization of basal portion of ventricular muscle near the atrioventricular ring.

'T' wave in ECG

- 'T' wave is due to the repolarization of ventricular musculature

No atrial repolarisation wave in ECG

- Atrial repolarization is not recorded as a separate wave in ECG because it merges with ventricular repolarization (QRS complex)

'P' wave is tall (more than 2.5 mm) in lead II and is usually pointed - Right atrial hypertrophy

P wave is tall and broad based or M shaped - Left atrial dilatation or hypertrophy

Small and shapeless 'P' wave, followed by a small compensatory pause - Atrial extrasystole

'P' wave is absent or small – Hyperkalemia/ Atrial fibrillation/ Middle AV nodal rhythm

'P' wave is inverted or absent - Sinoatrial block

'P' wave is inverted - Atrial paroxysmal tachycardia

'P' wave appears after QRS complex - Lower AV nodal rhythm

QRS is prolonged or deformed - Bundle branch block /Hyperkalemia

Hyperacute 'T' wave - Acute myocardial ischemia

'T' wave is small, flat or inverted - Old age, hyperventilation, anxiety, myocardial infarction, left ventricular hypertrophy and pericarditis

'T' wave is small, flat or inverted - Hypokalemia

'T' wave is tall and tented – Hyperkalemia

'U' wave in ECG

- It is supposed to be due to repolarization of papillary muscle

The sudden fall in blood pressure while standing for some time

- Orthostatic hypotension due to the effect of gravity with venous pooling

Respiratory System

Respiratory distress syndrome(RDS) or hyaline membrane disease in infants

- Deficiency of surfactant which reduces the surface tension of lungs

Under physiological conditions, the intrapleural pressure is always negative

- Intrapleural pressure is the pressure existing in pleural cavity, that is, in between the visceral and parietal layers of pleura. It is exerted by the suction of the fluid that lines the pleural cavity. This fluid is constantly pumped from the pleural cavity into the lymphatic vessels. Pumping of fluid creates the negative pressure in the pleural cavity.

The intrapleural pressure is responsible for venous return

- Because of the negative pressure in thoracic region, larger veins and vena cava are enlarged, i.e. dilated. Also, the negative pressure acts like suction pump and pulls the venous blood from lower part of body towards the heart against gravity. Thus, the intrapleural pressure is responsible for venous return. So, it is called the respiratory pump

Hyperventilation leads to dizziness, discomfort and chest pain

- Hyperventilation increases partial pressure of carbon dioxide ($p\text{CO}_2$) Excess of carbon dioxide stimulates the respiratory centers. During hyperventilation, excessive carbon dioxide is washed out. In blood, the partial pressure of carbon dioxide is reduced. It causes suppression of respiratory centers, resulting in apnea. Apnea is followed by Cheyne-Stokes type of periodic breathing. After a period of Cheyne-Stokes breathing, normal respiration is restored

Hypoventilation result in lethargy, coma and death

- Hypoventilation results in development of hypoxia along with hypercapnea. It increases the rate and force of respiration, leading to dyspnea

Reduced availability of oxygen to the tissues

- Hypoxia
 - Oxygen tension in arterial blood decreased
 - Oxygen carrying capacity of blood decreased
 - Velocity of blood flow decreased and
 - Utilization of oxygen by the cells lost

Bluish coloration of skin and mucus membrane

- It is due to the presence of large amount of reduced hemoglobin in the blood. Quantity of reduced hemoglobin should be at least 5 to 7 g/dL in the blood to cause cyanosis

Nervous System

Sensation complained by amputated patients in the missing limb

- Phantom limb pain (law of projection) - if a leg has been amputated, the cut end heals with scar formation. The cut ends of nerve fibers are merged within the scar. If the cut end of sensory fibers are stimulated during movement of thigh, the patient feels as if the sensation is originating from nonexistent leg

Cardiac pain is felt at inner part of left arm and left shoulder

- Referred pain is the pain that is perceived at a site adjacent to or away from the site of origin
According to dermatomal rule, pain is referred to a structure, which is developed from the same dermatome from which the pain producing structure is developed. A

dermatome includes all the structures or parts of the body, which are innervated by afferent nerve fibers of one dorsal root. ie, the heart and inner aspect of left arm originate from the same dermatome. So, the pain in heart is referred to left arm

Stiffness of muscles, static tremors and bradykinesia in parkinsonism

- Parkinson disease occurs due to lack of dopamine caused by damage of basal ganglia. It is mostly due to the destruction of substantia nigra and the nigrostriatal pathway

Special Senses

If a person enters a dim-lighted room (darkroom) from a bright-lighted area, he is blind for some time

- Dark adaption is the process by which the person is able to see the objects in dim light.
Dark adaptation is due to the increased sensitivity of rods as a result of resynthesis of rhodopsin and dilatation of pupil

When a person enters a bright-lighted area from a dim-lighted area, he feels discomfort due to the dazzling effect of bright light.

- Light adaptation is the process in which eyes get adapted to increased illumination. The causes of light adaptation are reduced sensitivity of rods and constriction of pupil

Myopic eye is corrected by using a biconcave lens

- Myopia is the eye defect characterized by the inability to see the distant object. It is otherwise called short sightedness because the person can see near objects clearly but not the distant objects. In this case the anteroposterior diameter of the eyeball is abnormally long. Therefore, the image is brought to focus a little in front of retina. Light rays, after coming to a focus, disperse again so, a blurred image is formed upon retina. Light rays are diverged by the concave lens before entering the eye

The eye defect characterized by the inability to see near object

- It is otherwise known as long sightedness because the person can see the distant objects clearly but not the near objects. It is also called hyperopia. Hypermetropia is due to decreased anteroposterior diameter of the eyeball. So, even though the refractive power of lens is normal, the light rays are not converged enough to form a clear image on retina, i.e. the light rays are brought to a focus behind retina. It causes a blurred image of near objects. Hypermetropia is corrected by using biconvex lens. Light rays are converged by convex lens before entering the eye.